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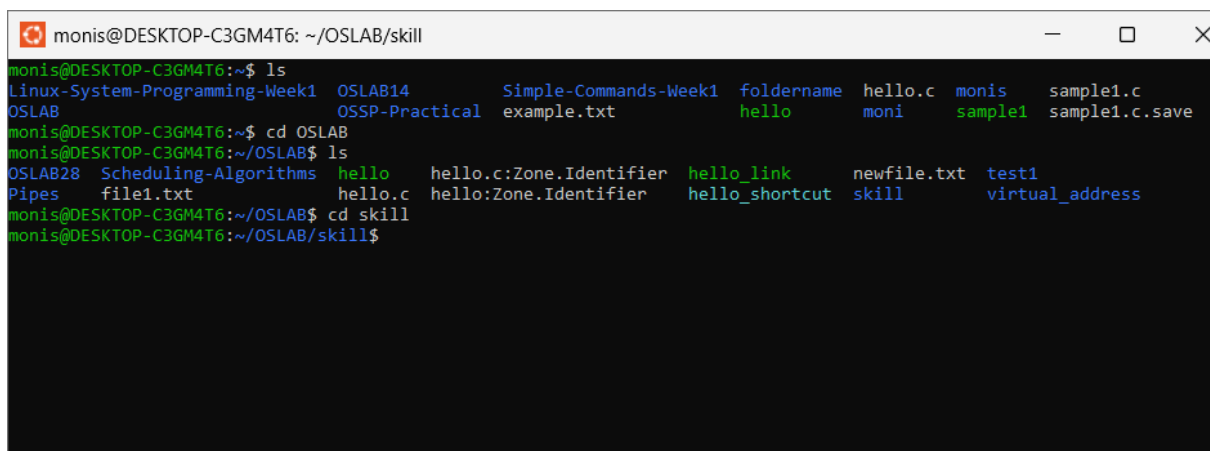
Section-6

OSSP (Operating Systems And Systems Programming) - 25CS2104E

(SKILL Content Delivery Concepts)

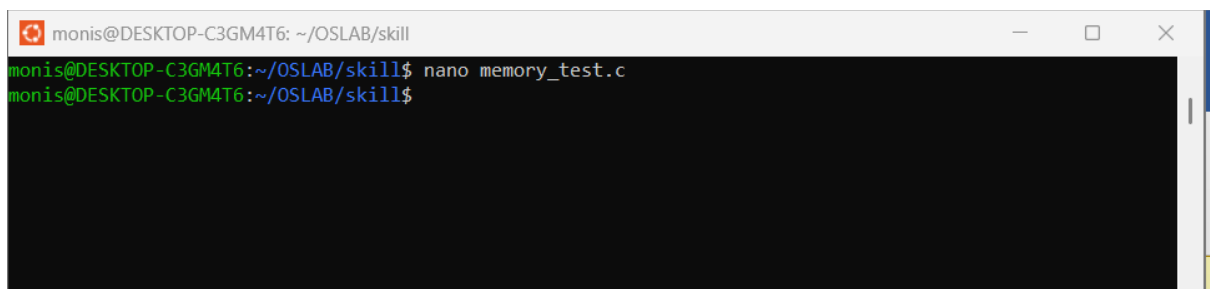
22.To Run Valgrind, Identify Leaks, Fix Memory Errors, Release Resources, Verify Cleanup, Document Findings.

To Design Test Cases, Create Test Scripts, Automate Execution, Verify Outputs, Compare Results, Generate Reports.



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~$ ls
Linux-System-Programming-Week1  OSLAB14      Simple-Commands-Week1  foldername  hello.c  monis  sample1.c
OSLAB                           OSSP-Practical  example.txt            hello       moni    sample1  sample1.c.save
monis@DESKTOP-C3GM4T6:~$ cd OSLAB
monis@DESKTOP-C3GM4T6:~/OSLAB$ ls
OSLAB28  Scheduling-Algorithms  hello      hello.c:Zone.Identifier  hello_link  newfile.txt  test1
Pipes    file1.txt              hello.c    hello:Zone.Identifier    hello_shortcut  skill        virtual_address
monis@DESKTOP-C3GM4T6:~/OSLAB$ cd skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 2:



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano memory_test.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 memory_test.c *
#include <stdio.h>
#include <stdlib.h>

int main(void)
{
    int *numbers = malloc(5 * sizeof(int));

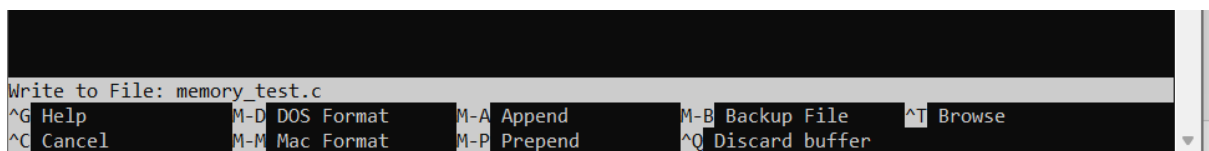
    if (numbers == NULL)
        return 1;

    for (int i = 0; i < 5; i++)
        numbers[i] = (i + 1) * 10;

    for (int i = 0; i < 5; i++)
        printf("%d ", numbers[i]);

    printf("\n");

    free(numbers);
    return 0;
}
```



Step 3 — Compile and run Valgrind

```
gcc -g -Wall -Wextra memory_test.c -o memory_test
```

```
valgrind --leak-check=full \
  --log-file=valgrind_report.txt \
  ./memory_test
```

```
cat valgrind_report.txt
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano memory_test.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ gcc -g -Wall -Wextra memory_test.c -o memory_test
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ valgrind --leak-check=full --log-file=valgrind_report.txt ./memory_test
10 20 30 40 50
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ cat valgrind_report.txt
==2029== Memcheck, a memory error detector
==2029== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==2029== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==2029== Command: ./memory_test
==2029== Parent PID: 1647
==2029==
==2029== HEAP SUMMARY:
==2029==   in use at exit: 0 bytes in 0 blocks
==2029==   total heap usage: 2 allocs, 2 frees, 1,044 bytes allocated
==2029==
==2029== All heap blocks were freed -- no leaks are possible
==2029==
==2029== For lists of detected and suppressed errors, rerun with: -s
==2029== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 1 — Create the expected output

Copy and paste:

`cd ~/OSLAB/skill`

`./memory_test > expected_output.txt`

```
Select monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./memory_test > expected_output.txt
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 2 — Create test.sh

`nano test.sh`

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./memory_test > expected_output.txt
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano test.sh
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 test.sh *
#!/bin/bash

echo "Q22 AUTOMATED TEST REPORT" > test_report.txt

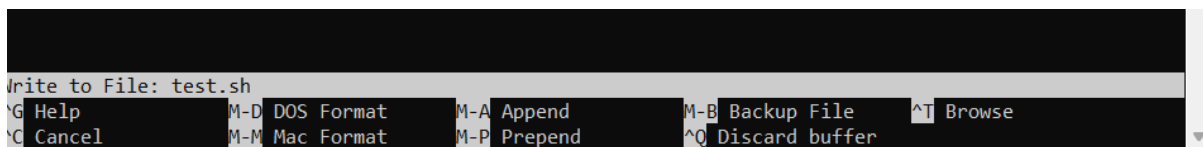
./memory_test > actual_output.txt

if diff -q expected_output.txt actual_output.txt; then
    echo "Output Test: PASS" | tee -a test_report.txt
else
    echo "Output Test: FAIL" | tee -a test_report.txt
fi

valgrind --leak-check=full \
    --error-exitcode=1 \
    --log-file=valgrind_report.txt \
    ./memory_test > /dev/null

if [ $? -eq 0 ]; then
    echo "Memory Test: PASS" | tee -a test_report.txt
else
    echo "Memory Test: FAIL" | tee -a test_report.txt
fi
```

Save using Ctrl + O, Enter, then Ctrl + X.



Step 3 — Run the automated tests

```
chmod +x test.sh
```

```
./test.sh
```

```
cat test_report.txt
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ chmod +x test.sh
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./test.sh
Output Test: PASS
./test.sh: line 24: syntax error: unexpected end of file from `if' command on line 18
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ cat test_report.txt
Q22 AUTOMATED TEST REPORT
Output Test: PASS
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```